

EMERGING TECHNOLOGY • HUMAN APPROVAL • CONTENT AUTOMATION

AI Tech Briefing System

A model-agnostic, human-approved pipeline that turns verified Notion records into UK-focused LinkedIn posts and carousel briefings.

CURRENT STATUS

Working local system with rendered outputs and manual publication evidence

EVIDENCE REVIEWED

Evidence basis: repository source and documentation, 26 commits dated 16-19 August 2026, generated PDF/PNG/caption outputs, and 131 tests rerun on 26 August 2026. Publication is recorded in the repository; the external LinkedIn post was not independently checked.

Problem

Technology news is abundant, but turning it into a consistent briefing requires more than summarisation. Claims need primary-source traceability, uncertainty must survive editing, UK relevance must be explicit, personal judgement must not be impersonated by an agent, and final artefacts need a reviewable publishing gate.

Objective

Build a repeatable content operation that selects eligible Notion records, validates their editorial fields, generates platform-ready assets and stops before publication unless both the source state and exact output files are still approved.

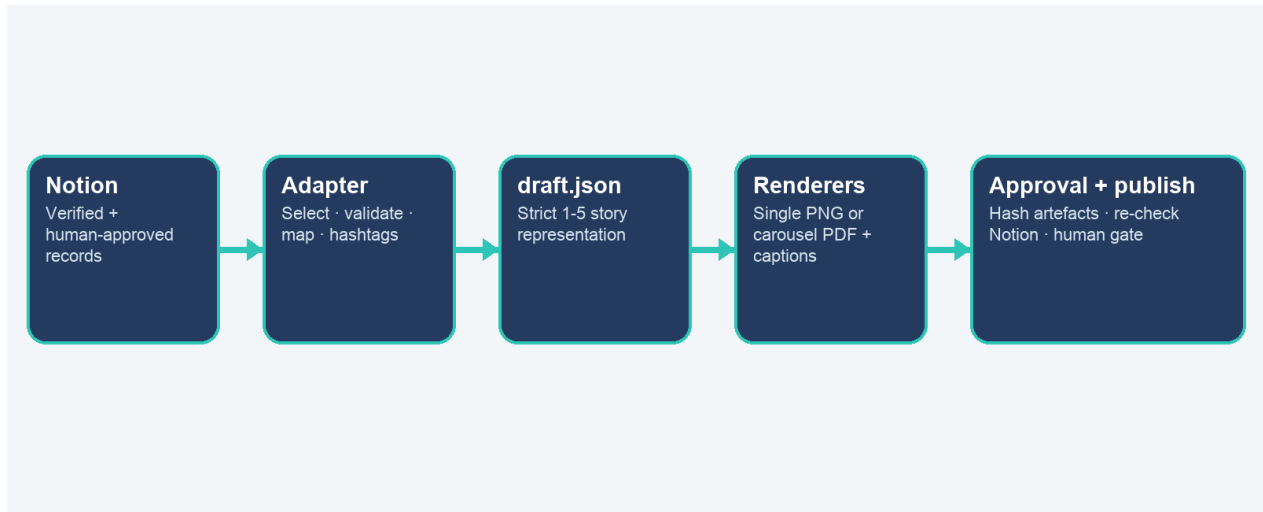
WHY THIS MATTERS

The project demonstrates emerging-technology communication as an implementation problem: data models, editorial policy, layout constraints, approval state and publishing controls must work together if AI-assisted content is to remain credible.

SYSTEM VIEW

Architecture and workflow

The active path begins with verified, human-approved Notion records. A mechanical adapter creates a strict 1-5 story draft; renderers produce a single image or carousel; approval hashes every artefact and re-checks Notion again before publication.



Architecture reconstructed from the reviewed implementation artefacts.

Operational workflow

1. Capture, verify and enrich candidate stories in the Notion intelligence database.
2. Select one daily story or assemble a weekly/legacy multi-story briefing under regional-mix rules.
3. Validate headline, summary, verified context, UK interpretation, personal take and practical action.
4. Generate content-aware hashtags and a canonical draft.json representation.
5. Render a 9:11.25 single-post PNG or variable-length 4:5 carousel PDF plus captions.
6. Review source URLs and layout; approve the exact hashes only while Notion records remain approved.
7. Publish manually or through the implemented LinkedIn API path with an explicit confirmation phrase.

FOUNDATIONS

Technologies and controls

The stack below reflects the reviewed implementation; legacy or optional paths are identified explicitly.

AREA	EVIDENCED TECHNOLOGY / PATTERN
Application	Python 3.11+, Pydantic 2, command-line interface
Rendering	ReportLab, PyMuPDF, pypdf test inspection, PNG/PDF outputs
Data and workflow	Notion API, JSON intermediate files, source registries
Inputs	RSS/Atom via feedparser in the legacy/offline collection path
Publishing	LinkedIn document/image publishing code with explicit confirmation
Quality	pytest, layout overflow checks, hash approval, Notion state re-verification
Model options	Command wrapper, Ollama, OpenAI-compatible, Anthropic and Gemini adapters in the legacy curation path

DELIVERY

What was actually built

Evidence labels separate implemented work from claims, plans and dependencies.

IMPLEMENTED Notion-to-briefing adapter

Maps approved records into a strict briefing schema without silently rewriting or truncating editorial copy.

IMPLEMENTED Daily and weekly selection paths

Supports one focused post per posting day and a Sunday digest, while retaining the original five-story carousel path.

IMPLEMENTED Fact/opinion separation

Verified context, UK interpretation, personal judgement and practical action are distinct fields with different ownership and constraints.

IMPLEMENTED Single-post and carousel renderers

Produces LinkedIn-ready PNG and PDF assets with pre-flight fit checks and source lines.

IMPLEMENTED Unified approval manifest

Hashes the draft, PDF, captions and any standalone posts; live Notion state is checked at approval and again before publishing.

IMPLEMENTED Publishing boundary

A LinkedIn publisher and dry-run exist, but unattended live publishing is deliberately not enabled.

IMPLEMENTED Operational evidence

Repository outputs include three eight-page briefings and a later single-story image/caption; the first post is recorded as manually published.

Implementation notes

- The adapter is intentionally mechanical: invalid or oversized copy fails with a named record rather than being silently shortened.
- Five-story briefings enforce a 3 UK + 2 global split, with a documented 4 + 1 fallback; smaller daily/weekly drafts use a rolling target at selection time.
- The mandatory 'My Take' content is treated as the operator's judgement; agents may draft candidates but cannot finalise it as the user's voice.
- Page fit is calculated from interacting headline, summary and callout heights, not only individual field-length limits.
- The older feed/model curation components remain for offline fixtures; the production-oriented path reads from Notion.

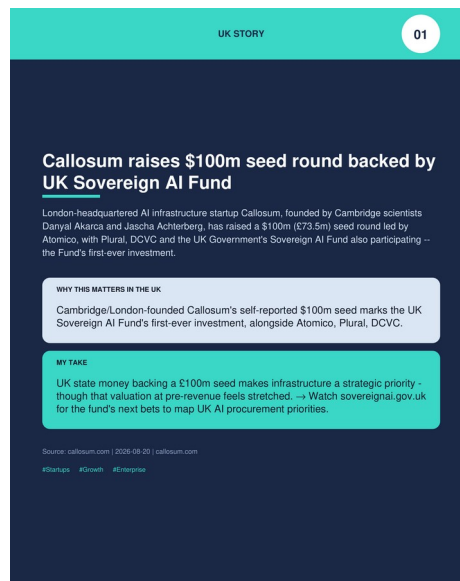
ASSURANCE

Testing and validation evidence

Checks were rerun where the local repository and dependencies made that practical.

CHECK	RESULT	EVIDENCE
Automated suite	PASS	131 tests passed on 26 August 2026 in 3.41 seconds; warnings were dependency deprecations.
Coverage areas	PRESENT	Models, providers, adapter, Notion store, renderer, social posts, approval, publisher, registry and test isolation are represented.
Layout safety	PASS	Regression tests cover production-length headlines/summaries, actual renderer fit prediction and explicit overflow failure.
Approval integrity	PASS	Tests detect changed standalone assets and Notion approval drift after local approval.
Output evidence	PRESENT	Three repository PDF briefings contain eight pages each; a later rendered single-post PNG and caption are also present.
Publication evidence	REPOSITORY-RECORDED	Commits and notes record a first manual publication. No independent LinkedIn/API verification was performed for this review.

Example output



Example single-post asset found in repository output. Included as implementation evidence; the underlying news claim was not re-verified during this portfolio review.

REFLECTION

Limits, lessons and next steps

Limitations

- Requires a correctly configured Notion database and credentials; production behaviour depends on that external state.
- Human source checking remains mandatory; automated schema and layout tests do not prove every news claim is accurate.
- LinkedIn API publishing is implemented but unattended live publishing is intentionally disabled.
- The repository's recorded first publication was not independently verified against the live social post.
- The cadence has a planned graduation from three posts per week to daily only after several stable weeks; that maturity threshold is not evidenced as complete.
- Legacy and active paths increase maintenance surface and require clear documentation to avoid using the wrong workflow.

Lessons learned

- Verified facts, editorial interpretation and personal judgement need separate data fields and ownership rules.
- Approval must cover both content state and immutable artefacts; checking only one leaves a drift gap.
- Real layout is a system constraint: passing text-length validation does not guarantee that a page will render.
- A weak news day should stop the pipeline instead of lowering the evidence bar.

Next steps

1. Operate the three-times-weekly cadence long enough to assess quality corrections before increasing frequency.
2. Add auditable publication receipts or API response evidence without storing access tokens.
3. Reduce legacy-path complexity once fixture and migration needs are settled.
4. Add end-to-end tests against a controlled Notion sandbox and LinkedIn dry-run fixtures.
5. Continue documenting corrections and source-verification decisions after publication.

Evidence base reviewed

- README.md, docs/operating-contract.md and docs/editorial-policy.md
- pyproject.toml and config source registries
- src/intelligence/* and src/briefing/*
- tests/*.py (131 passing in current rerun)
- output/pdf/*, output/social-posts/* and generated captions
- Commits 37f9208 through 44bb2ee

CLAIM DISCIPLINE

Implemented features are limited to artefacts directly supported by the reviewed repository snapshot. Planned, experimental, recorded or externally dependent items are labelled accordingly.